Ideation Canvas

Project: AgroVerify | Domain: Agriculture | Problem: Counterfeit Inputs | Team: Agri_Developers

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People

Primary Users

- Small & Marginal Farmer (86% of Indian farmers, <2 hectares) — buys seeds, fertilizers, pesticides before every sowing season; has no verification tool; loses entire season if inputs are fake
- Women Farmers — perform significant farm labor; often make input purchases; equally excluded from verification systems

Secondary Users

- Agri-input Dealers — sell inputs; mostly unaware when products are fake; fear being reported; need a compliance dashboard
- Krishi Vigyan Kendra (KVK) Officers — visit villages periodically; can champion app adoption; need data on fake-input hotspots

Tertiary Users

- State Agriculture Department Officers — need real-time data on where fake inputs circulate to enable enforcement action at district level
- CSC (Common Service Centre) Operators — can assist farmers with app use; already present in most villages as digital access points

Activities

What the User Does Today (Without Solution)

- Visits 1-3 input shops before sowing; compares prices only — never verifies certification
- Buys inputs on dealer recommendation and brand name familiarity
- Sows crop; waits 3-4 weeks; discovers germination failure when it is too late
- Has no recourse — dealer denies fake product; no receipt; no batch record

What the User Will Do With AgroVerify

- Picks up packet → opens AgroVerify → scans QR OR dials toll-free number
- Receives instant result in local language: GENUINE ✓ or SUSPECT ✗ with details
- Decides whether to buy based on verified information — not dealer trust
- Suspicious detections auto-logged to district agriculture officer dashboard

Frequency

- 2-3 times per year per farmer (Kharif + Rabi seasons) — habitual use builds quickly

Situation / Context / Location

What — The Core Problem Situation

- ~25-30% of India's pesticide market is fake/substandard. 1-in-8 seed batches tested in 2024-25 were substandard. Farmers have no tool to verify before buying.

When — Critical Timing

- 2-4 weeks before Kharif (June) and Rabi (Nov) sowing seasons — farmer has cash and urgency
- Purchase decision takes 10-15 minutes — verification must fit within this window

Why — Root Cause

- SATHI (govt seed tracking) stops at dealer/producer level — no farmer-facing interface built
- No legal mandate for dealers to show verification; no penalty framework at point-of-sale

Where — Location

- Rural agri-input shops — often in village center or block-level town
- Low internet connectivity; often 2G or no data — solution must work offline or on SMS
- Farmer may be 10-30 km from nearest KVK office or agriculture authority

Props / Tools / Objects / Equipment

Tools Farmer Has

- Basic feature phone (₹1,500-3,000) OR low-end Android (₹4,000-8,000) — AgroVerify must work on both
- USSD/SMS capability — fallback for zero-internet use: farmer dials *123*BATCHCODE# and gets SMS reply with verification result

Tools AgroVerify Uses

- SATHI Government API — batch-level seed/pesticide/fertilizer data; AgroVerify reads and caches this data
- QR Code Scanner — lightweight, works in low-light conditions of a shop interior
- Offline cache — syncs when connected; allows verification in no-internet zones using last-known batch data
- Local Language NLP — result shown in Gujarati, Hindi, or Marathi based on phone locale

Infrastructure Used (Already Exists)

- Common Service Centres (CSCs) — 5+ lakh across India; can assist farmers with app use and onboarding
- Krishi Vigyan Kendras — 731 centres nationally; can train farmers and distribute app via WhatsApp/SMS

What Is New / Innovated

- Farmer-facing last-meter extension of SATHI — no new database built; leverages existing government data
- Dealer Compliance Dashboard — agriculture officers see real-time map of fake-input detections by village
- Crowdsourced Alert Network — if 3+ farmers flag same batch as suspect, auto-alert sent to district officer